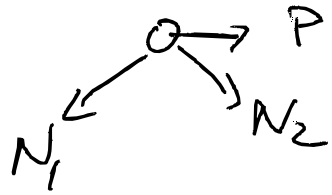
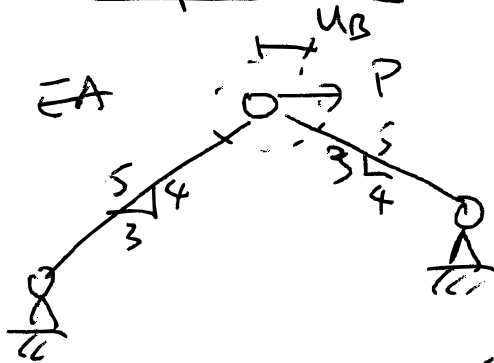


Example 1



$$\sum F_x = 0 \Rightarrow \frac{4}{5} N_2 + P - \frac{3}{5} N_1 = 0$$

$$\sum F_y = 0 \Rightarrow -\frac{4}{5} N_1 - \frac{3}{5} N_2 = 0$$

$$\Rightarrow N_1 = \frac{3}{5} P, N_2 = -\frac{4}{5} P$$

$$W = \frac{1}{2} P u_B$$

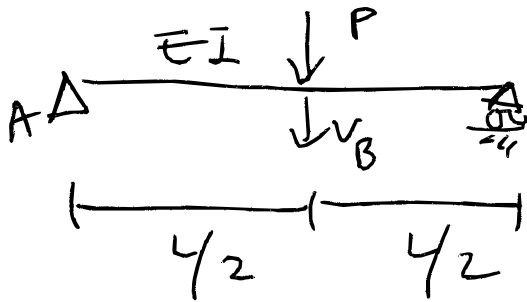
$$U = \sum \frac{1}{2} \frac{N^2 L}{EA}$$

$$= \frac{1}{2} \left(\frac{N_1^2 L}{EA} + \frac{N_2^2 L}{EA} \right), L=5$$

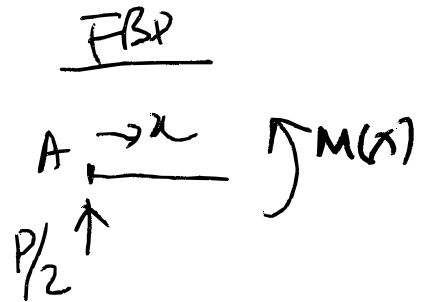
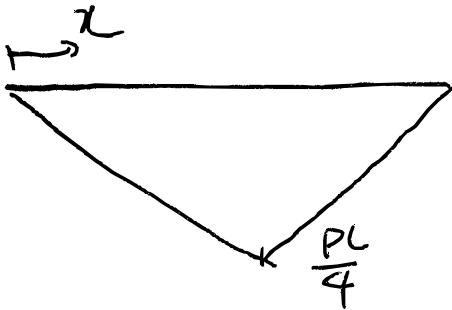
$$= \frac{1}{2EA} 5P^2$$

$$\text{Since } W = U \Rightarrow u_B = \frac{5P}{EA}$$

Example 2



$$W = \frac{1}{2} P v_B$$



$$U = \int_0^L \frac{M^2(x)}{2EI} dx = 2 \int_0^{L/2} \frac{M^2}{2EI} dx$$

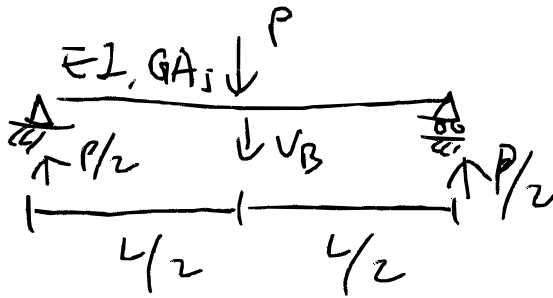
$$= 2 \int_0^{L/2} \left(\frac{P}{2} x \right)^2 \frac{1}{2EI} dx$$

$$= \frac{P^2 L^3}{96EI}$$

$$W = U \Rightarrow \frac{1}{2} P v_B = \frac{P^2 L^3}{96EI}$$

$$\Rightarrow v_B = \frac{PL^3}{48EI}$$

Example 3

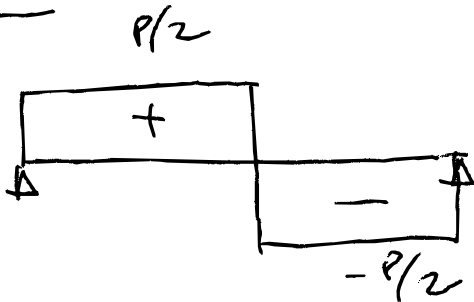


$$W = \frac{1}{2} P V_B$$

$$U = \underbrace{U_M}_{\text{bending}} + \underbrace{U_V}_{\text{shear}}$$

From Example 2, $U_M = \frac{P^2 L^3}{96 EI}$

SFD



$$U_V = \int_0^L \frac{V^2}{2GA_s} dx$$

$$= 2 \int_0^{L/2} \frac{V^2}{2GA_s} dx$$

$$U_V = \frac{1}{GA_s} \int_0^{L/2} \left(\frac{P}{2}\right)^2 dx = \frac{P^2 L}{8GA_s}$$

$$\therefore W = U$$

$$\frac{1}{2} P V_B = \frac{P^2 L^3}{96 EI} + \frac{P^2 L}{8GA_s}$$

$$\Rightarrow V_B = \frac{PL^3}{48 EI} + \frac{PL}{4GA_s}$$

$$e = \frac{V_{B, \text{shear}}}{V_{B, \text{bending}}} = \frac{PL/4GA_s}{PL^3/48EI}$$

$$= \frac{12EI}{GA_s L^2}$$

Poisson's ratio $\nu \sim 0.2 - 0.3$

$$G = \frac{E}{2(1+\nu)} \sim 0.4E$$

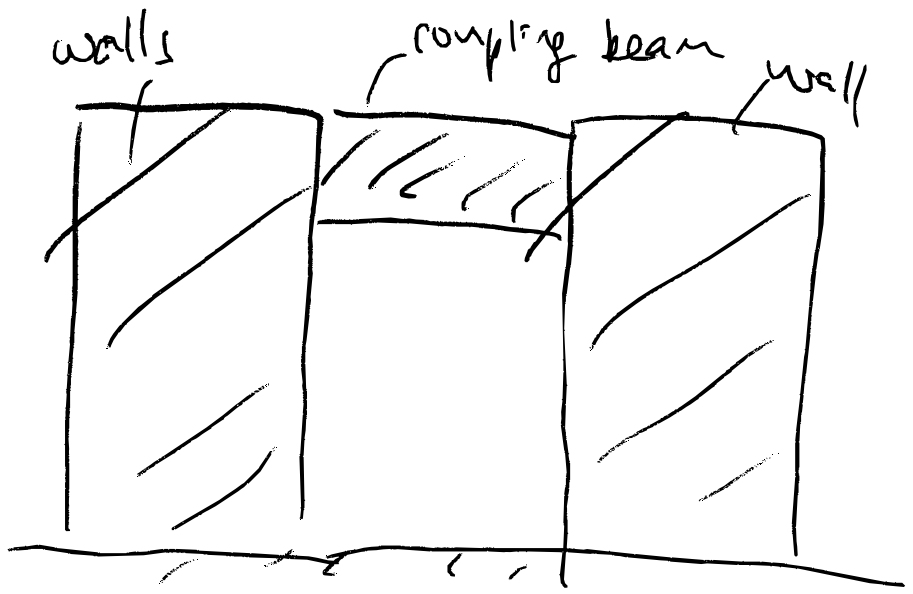
$$\frac{I}{A_s L^2} \sim \frac{1}{[L/(I/A)]^2}$$

(radius of gyration)²

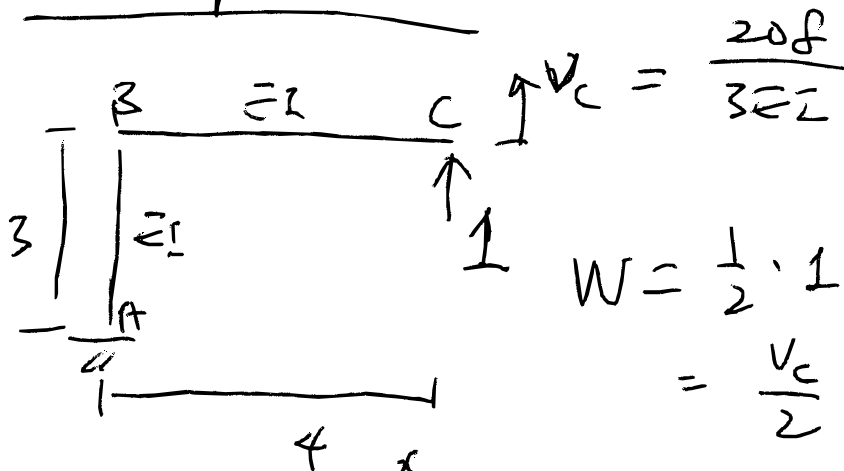
$$e \sim 2.5 \left(\frac{d}{L}\right)^2, \quad d: \text{beam depth}$$

$$\sim \frac{2.5}{(L/d)^2}$$

L/d	10	5	2
e	2.5%	10%	62.5%



Example 4

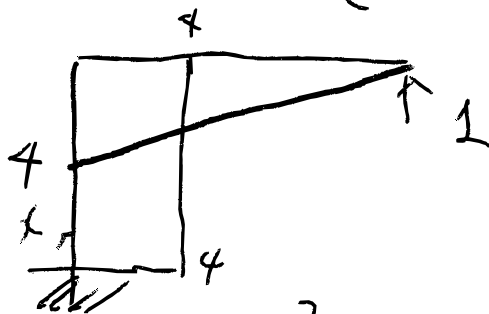


$$V_c = \frac{20f}{3EI}$$

$$W = \frac{1}{2} \cdot 1 \cdot V_c$$

$$= \frac{V_c}{2}$$

$$U = U_{col} + U_{beam}$$



$$U_{col} = \int_0^3 \frac{M^2}{2EI} dx = \int_0^3 \frac{4^2}{2EI} dx$$

$$= \frac{24}{EI}$$

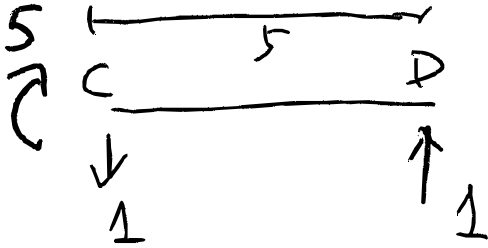
$$U_{beam} = \int_0^4 \frac{M^2}{2EI} dx = \int_0^4 \frac{x^2}{2EI} dx$$

$$= \frac{32}{3EI}$$

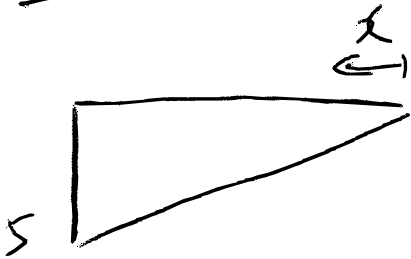
$$W = U_{col} + U_{beam} \Rightarrow V_c = \frac{20f}{3EI}$$

Example 5

Member CD



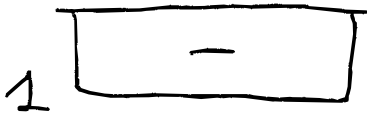
BMD



$$U_m = \int_0^5 \frac{m^2}{2EI} dx$$

$m = x$

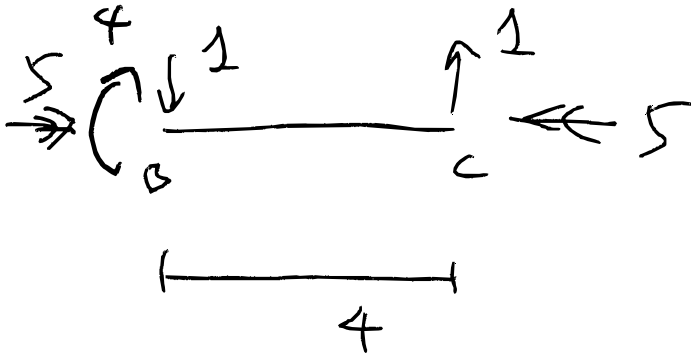
SFD



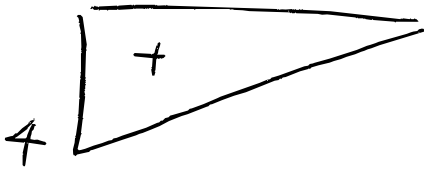
$$U_v = \int_0^5 \frac{v^2}{2GA_s} dx$$

$v = -1$

Member BC



BMD



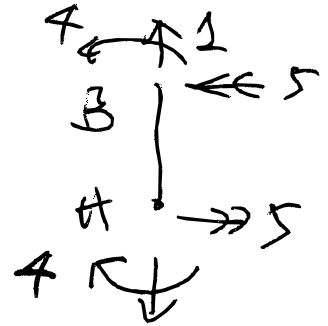
SFD



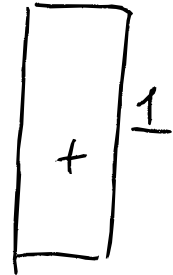
TMD



Member AB



AFD



BMD

